1. Many good examples of "selection bias" or "survivorship bias" were suggested during the tutorial sessions. The following examples are especially close to the "return-to-education" example:

$$investment_i = \beta_0 + \beta_1 experience_i + u_i, \quad i = 1, ..., n. \tag{1}$$

where $investment_i$ refers to person i's investment gain in the financial market, $experience_i$ refers to years of investment experience, and the error term u_i includes i's innate ability. Therefore, u_i will be positively correlated with the explanatory variable $experience_i$ as those with higher innate ability are more likely to survive in the market and gain years of experiences. Similarly,

$$health_i = \beta_0 + \beta_1 gym_i + u_i, \quad i = 1, ..., n. \tag{2}$$

where $health_i$ refers to person i's health condition, gym_i refers to hours of gym exercises, and the error term u_i includes i's innate ability. Here, u_i may be positively correlated with gym_i because on average, people with higher innate physical ability (or mental ability such as endurance) are more likely to survive hard exercises and continue the training. **This implies that in the example of** *health* **and** *gym***, when the value of the regressor** X **(***gym***) increases, the value of** $E[u|X]$ **also tend to increase. Therefore, the OLS Assumption 1 will be violated.**

2. Figure 1(a) shows the **relationship between the error term** u **and the explanatory variable** X. When OLS Assumption 1 holds, $E[u_i|X_i = 5] = 0$. Therefore, at the point "5", some error terms are positive (+), and some error terms are negative (-). But the average of these error terms is equal to zero at the point "5". Similarly, OLS Assumption 1 also requires that $E[u_i|X_i = 10] = 0, E[u_i|X_i = 15] = 0, ...$ That is, the average of the error terms are also equal to zero at the points "10", "15" and other points of X.

   Figure 1(b) shows the **relationship between the dependent variable** Y **and the explanatory variable** X. This figure is obtained by combining Figure (a) with the population regression line $\beta_0 + \beta_1 X$ (Note that each Y_i can be obtained by combining **the unsystematic part** u_i **and the systematic part** $\beta_0 + \beta_1 X_i$). As can be seen in Figure (b), at the point "5", some values of Y_i are above the population regression line, and some values of Y_i are below the population regression line. But the average of these Y_i has to be exactly on the population regression line (according to OLS Assumption 1). Similar requirement for Y_i at the points "10", "15", etc.

   Figure 1(c) has the same meaning as Figure (a). The only difference is that Figure (c) replaces the data points in Figure (a) with the **distributions of** u_i at $X_i = 5, 10, 15$.

Notice that for each value of X_i, the density curve of u_i is highest at zero, which means that the data point of u_i are most likely to be observed around zero.

Figure 1(d) has the same meaning as Figure (b). The only difference is that Figure (d) replaces the data points in Figure (b) with the **distributions of** Y_i at $X_i = 5, 10, 15$.

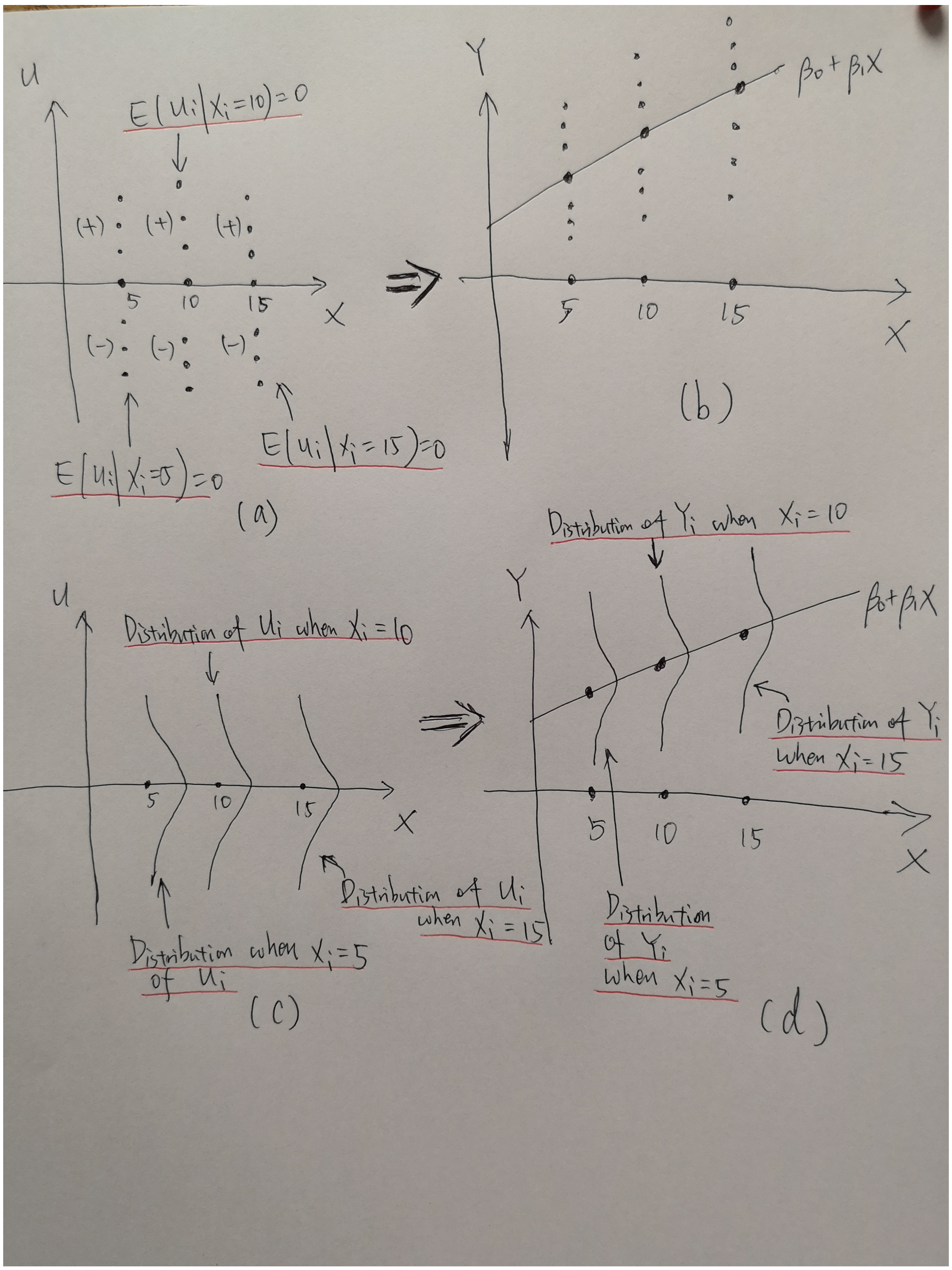

Figure 1

To give an example where $E[u] = 0$ but $E[u|X] \neq 0$, for simplicity we can consider the case where X only takes two values such as 0 and 1. Then, we can show in the diagram that even if $E[u|X = 0]$ is negative while $E[u|X = 1]$ is positive, it is still possible for $E[u]$ to be equal to zero as it is the overall average over the two cases (both $X = 1$ and $X = 0$).

3. (a) We may define the following linear regression model:

$$Y_i = \beta_0 + \beta_1 X_i + u_i, \quad i = 1, ..., n, \tag{3}$$

where Y_i is the wage of the graduate i, X_i is the binary regressor with $X_i = 1$ if i graduated from SSS and $X_i = 0$ if i graduated from SOH, and u_i is the error term, which includes all the other factors that may affect i's wage.

(b) The OLS Assumption 1 can be written as

$$E[u|X] = E[\,\text{innate ability}\,|\,\text{school type}\,] = 0. \tag{4}$$

In particular, this implies that $E[\,\text{innate ability}\,|\,\text{SSS}\,] = E[\,\text{innate ability}\,|\,\text{SOH}\,]$, that is, the average ability of SSS graduates is the same as that of SOH graduates. For example, if the students are randomly admitted to the two programs (SSS and SOH), then we can expect that the OLS Assumption 1 to hold. However, in practice the admission is not based on random assignment (indeed, the selection criteria may be very different for the two schools), so that there may exist difference in average abilities between SSS and SOH.

(c) By the derivations in the lecture note, we have in general,

$$\begin{aligned} E[Y_i|X_i] &= E[\beta_0 + \beta_1 X_i + u_i|X_i] \\ &= E[\beta_0|X_i] + E[\beta_1 X_i|X_i] + E[u_i|X_i] \\ &= \beta_0 + \beta_1 E[X_i|X_i] + E[u_i|X_i] \\ &= \beta_0 + \beta_1 X_i, \end{aligned} \tag{5}$$

where $E[u_i|X_i] = 0$ by OLS Assumption 1. Then, when $X_i = 1$ or 0, we have

$$E[Y_i|X_i = 1] = \beta_0 + \beta_1, \tag{6}$$
$$E[Y_i|X_i = 0] = \beta_0. \tag{7}$$

Therefore, β_0 corresponds to the average wage of the graduates from SOH, and

$$\beta_1 = E[Y_i|X_i = 1] - E[Y_i|X_i = 0], \tag{8}$$

which corresponds to the difference in average wages between graduates from SSS and SOH. Notice that this is a model with binary regressors, so β_1 is **not interpreted as a slope or rate of change**.

(d) Refer to Figure 2.

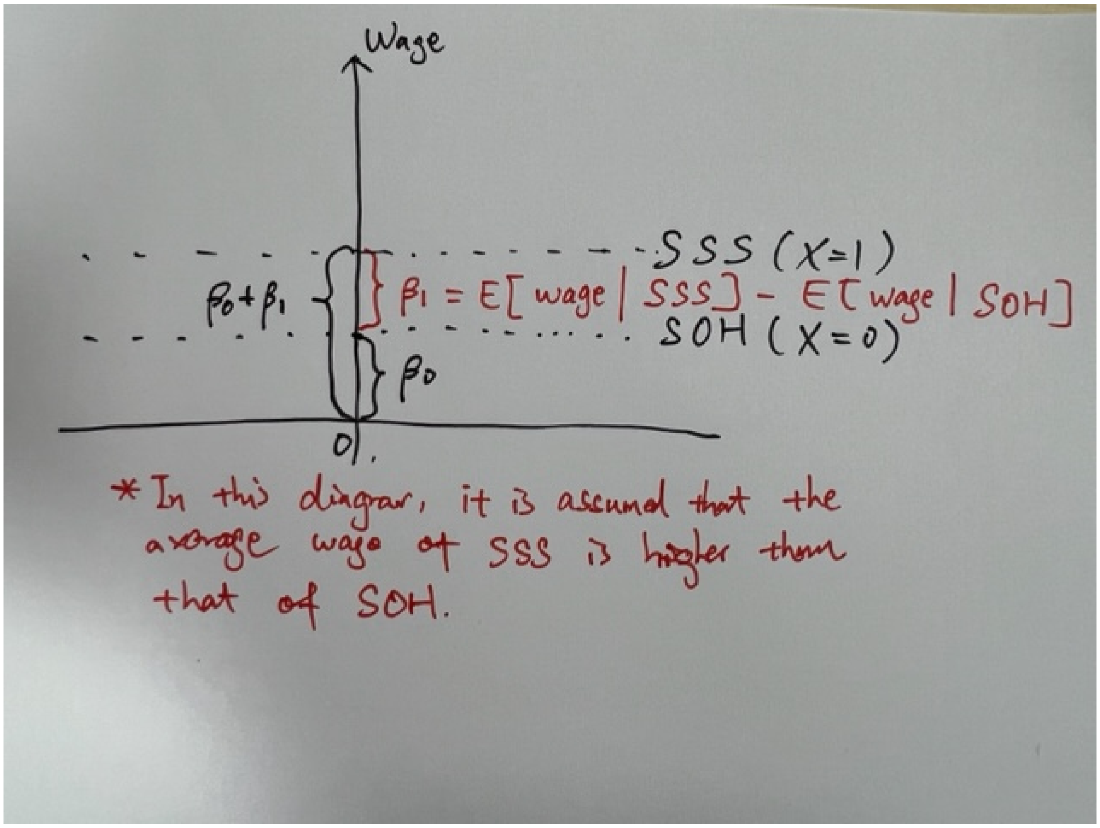

Figure 2

4. In general, for $\tilde{X}_i = c_1 X_i$ and $\tilde{Y}_i = c_2 Y_i$, where c_1 and c_2 are constants ($c_1 \neq 0$), the following equation holds for the sample average of $\tilde{X}_i, i = 1, ..., n$ (denoted as $\bar{\tilde{X}}$):

$$\bar{\tilde{X}} = \frac{1}{n}\sum_{i=1}^{n}(c_1 X_i) = c_1\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right) = c_1 \bar{X}, \tag{9}$$

and the following equation holds for the sample average of $\tilde{Y}_i, i = 1, ..., n$ (denoted as $\bar{\tilde{Y}}$):

$$\bar{\tilde{Y}} = \frac{1}{n}\sum_{i=1}^{n}(c_2 Y_i) = c_2\left(\frac{1}{n}\sum_{i=1}^{n} Y_i\right) = c_2 \bar{Y}. \tag{10}$$

Therefore, for the OLS estimator $\tilde{\beta}_1$, we have:

$$\begin{aligned}
\tilde{\beta}_1 &= \frac{\sum_{i=1}^{n}(\tilde{X}_i - \bar{\tilde{X}})(\tilde{Y}_i - \bar{\tilde{Y}})}{\sum_{i=1}^{n}(\tilde{X}_i - \bar{\tilde{X}})^2} & (11)\\
&= \frac{\sum_{i=1}^{n}(c_1 X_i - c_1\bar{X})(c_2 Y_i - c_2\bar{Y})}{\sum_{i=1}^{n}(c_1 X_i - c_1\bar{X})^2} & (12)\\
&= \frac{\sum_{i=1}^{n} c_1 c_2 (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^{n} c_1^2 (X_i - \bar{X})^2} & (13)\\
&= \frac{c_1 c_2 \sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})}{c_1^2 \sum_{i=1}^{n}(X_i - \bar{X})^2} & (14)\\
&= \left(\frac{c_2}{c_1}\right)\frac{\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^{n}(X_i - \bar{X})^2} = \left(\frac{c_2}{c_1}\right)\hat{\beta}_1. & (15)
\end{aligned}$$

In the current question, $c_1 = c_2 = 10$ since $\tilde{X}_i = 10 \cdot X_i$ and $\tilde{Y}_i = 10 \cdot Y_i$, therefore $\tilde{\beta}_1 = \hat{\beta}_1$. Then, for $\tilde{\beta}_0$ we have:

$$\begin{aligned}
\tilde{\beta}_0 &= \bar{\tilde{Y}} - \tilde{\beta}_1 \bar{\tilde{X}} && (16)\\
&= c_2 \bar{Y} - \tilde{\beta}_1 (c_1 \bar{X}) && (17)\\
&= c_2 \bar{Y} - \left(\frac{c_2}{c_1}\right) \hat{\beta}_1 \cdot \left(c_1 \bar{X}\right) && (18)\\
&= c_2 \bar{Y} - c_2 \hat{\beta}_1 \bar{X} = c_2 \hat{\beta}_0, && (19)
\end{aligned}$$

where the second equality follows from (9) and (10), the third equality follows from $\tilde{\beta}_1 = \left(\frac{c_2}{c_1}\right) \hat{\beta}_1$, and the last equality follows from $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$. As $c_2 = 10$ in the current question, we obtain $\tilde{\beta}_0 = 10\hat{\beta}_0$.